

Table 1: Evaluation Results on Uniform-MCF-500, Uniform-MCF-5k and Uniform-MCF-10k. We compared our matrix-vector iterative algorithm (EODT-Sinkhorn) with classical MCF algorithms (such as Real and ZKW) and commonly used general solver Gurobi. One can find that our algorithm is significantly more efficient than traditional algorithms when computing large-scale Maximum Flow instances, all while maintaining low computational errors. The fastest solution time is bolded.

Size	Methods	w/o constraint obj	w/o constraint time	w/ edge constraint obj	w/ edge constraint time	w/ node constraint obj	w/ node constraint time	w/ edge+node constraints obj	w/ edge+node constraints time
500×500	Real	10.02	14.8 s	10.067	101 s	10.065	101 s	10.066	101.9s
	ZKW	10.02	18.4 s	10.067	129 s	10.065	139.4 s	10.066	120.8 s
	Gurobi	10.02	768 s	10.067	4254 s	10.066	4314 s	10.067	4052 s
	ours	10.11	4.18 s	10.137	34 s	10.67	43.4 s	10.138	72.8 s
$5k \times 5k$	Real	10.059	1843 s	10.071	1036 s	10.071	899 s	10.064	868 s
	ZKW	10.059	2034 s	10.071	977s	10.071	1200 s	10.064	1015 s
	Gurobi	-	≥ 3 hours	-	≥ 3 hours	-	≥ 3 hours	-	≥ 3 hours
	ours	10.311	33 s	10.18	814 s	10.431	316 s	10.186	656 s
$10k \times 10k$	Real	10.08	3363 s	10.0898	2032 s	10.0898	1842 s	10.0898	1425 s
	ZKW	10.08	3781 s	10.0898	2396 s	10.0898	2543 s	10.0898	1961 s
	Gurobi	-	≥ 3 hours	-	≥ 3 hours	-	≥ 3 hours	-	≥ 3 hour
	ours	10.728	47 s	10.1496	1352 s	10.18	1021 s	10.147	1462 s

- **NETGEN (Integer-precision)**: Note that NETGEN (Klingman et al., 1974) is a well-established generator that produces integer-precision instances for the Minimum Cost Flow (MCF) problem and other network optimization problems. It is designed to generate minimum-cost flow problems based on a set of parameters including node size, arc costs, and arc capacities. In our data generation process, we utilize PyNETGEN , a Python implementation of NETGEN. For further details, please refer to Appendix D.
- **Vision (Real Scene)**: This family(Goldberg, 2008) consists of MCF instances based on large-scale maximum flow problems arising in computer vision applications(with over one million nodes). These data files were made available by the Computer Vision Research Group at the University of Western Ontario. The detail of the data is given in Appendix D.

Baselines. To demonstrate the feasibility and effectiveness of our OFT-Sinkhorn method, we compare it with representative MCF algorithms (including classic and SOTA algorithms). We consider the following five methods as our baselines: **1)Real((Papadimitriou & Steiglitz))** is an exact algorithm that combines SPFA (Ahuja et al., 1995) and augmenting paths which is particularly suitable for solving minimum cost flow problems in graphs of fixed distribution of capacity; **2)ZKW((Goldberg et al.))** is a variant of Successive Shortest Path Algorithm based on augmenting paths and DFS to efficiently find the shortest path. It is particularly suitable for solving minimum cost flow problems in graphs of random distribution of capacity. **3)Gurobi((Gurobi Optimization, LLC, 2021))** is a powerful optimization solver that can efficiently handle a wide range of mathematical optimization problems, including the minimum cost flow problem. However, as the number of nodes and instances increases, Gurobi’s computational efficiency decreases. **4)pns((Kara & Özturan, 2022))** is one of the SOTA Parallel network simplex algorithm for the MCF problem. **5)lemon((Király & Kovács, 2012))** is a Highly optimized CPU based solver for MCF problems. Here we test on lemon-ns which is the network simplex version of lemon.

4.2 EXPERIMENTAL RESULTS

In this section, we discuss the overall performance of various methods on uniform , NETGEN and Vision datasets which is presented in Table 1 , Table 2 and Table 3.

Evaluation on Uniform Data. Uniform-MCF is a double-precision MCF instance with predefined edge capacities. In terms of Obj across each dataset, both Real and ZKW consistently achieve the optimal solution, while Gurobi gave a tight approximate solution with a gap of less than 0.1% as shown in Table 1. This might be attributed to the fact that the MCF problem is solvable in polynomial time, where exact algorithms can always provide the optimal solution. However, their efficiency significantly decreases with larger datasets. Notably, Gurobi struggles with problems exceeding